1 Luminosity Calculation

In pp collision data, the analysis of rare probes uses selective jet and photon triggers. Thus, when calculating invariant cross sections, one must calculate the corresponding integrated luminosity $\mathscr{L}_{\text{int}}$. This section details this calculation including the accounting for pileup, i.e., multiple pp collisions in the same bunch crossing.

1.1 Getting the Basic Information

To calculate $\mathscr{L}_{\text{int}}$, it is critical that the value corresponds to the exact good run list (GRL) and segments analyzed for the yields, i.e., the extracted number of jets and photons. The calculation starts from the DST files. For each run, one iterates over all segments of the DSTs used in the analysis.

For each segment, there are nine values extracted for each trigger as listed below:

1. The raw scalers at the beginning of the segment
2. The raw scalers at the end of the segment
3. The live scalers at the beginning of the segment
4. The live scalers at the end of the segment
5. The scaled scalers at the beginning of the segment
6. The scaled scalers at the end of the segment
7. The number of triggers fired for each trigger in all **produced** events
8. The start BCO (this is a single number for the entire segment)
9. The end BCO (a single number for the entire segment)

Then one sums these values over the entire run (typically 60 minutes or less of data taking) and calculates the average prescale for each trigger for the whole segment. The average prescale is defined by the total live scalers / total scaled scalers over the entire run. The sampled number of "minimum bias" MBD collisions is then the summed live **MBDN&S $\geq$ 1** triggers divided by this average prescale. It is notable that in the sPHENIX DAQ, each trigger has a "scaledown" and a value of zero implies recording every live event. Thus the prescale referred to here has the relation prescale = scaledown + 1. If the physics channel, e.g., jet, does not require an **MBDN&S $\geq$ 1** coincidence condition, then one simply divides the above value by the fraction of the pp 42 mb inelastic cross section firing the **MBDN&S $\geq$ 1** . There is an important additional correction due to the finite probability of more than one pp collision occurring in a single bunch crossing, referred to as pile-up. The next section details this pile-up correction.

1.2 Correcting for Pileup

To correct for pileup, the key assumption is that all collisions are completely uncorrelated and thus follow a Poisson distribution,

$$P(n, \lambda) = \frac{e^{-\lambda}\lambda^n}{n!} \tag{1}$$

where $P(n)$ is the probability of n events in a given time interval. This uncorrelated assumption implicitly assumes that all bunches have the same intensity, which is a reasonable approximation. In this Poisson distribution equation, the parameter λ is the mean number of events. For this case, the mean number of collisions in a given bunch crossing is typically labeled μ (instead of λ).

$$\mu = \frac{R_{true}}{f}, \tag{2}$$

where f is the bunch crossing frequency = 111 bunches $\times$ 78.196 kHz = 8.86 MHz.

Two things are needed to calculate the multi-interaction corrected luminosity: a correction factor that upweights multi-interaction bunch crossings by the number of interactions that occur in the crossing, and a mapping from R^{raw}_{MBD}, the **MBDN&S $\geq$ 1** raw rate, to R_{true}, the true collision rate. The latter is necessary because R_{true} is not known, only R^{raw}_{MBD} is. The correction factor (F_{corr}) is simply

$$F_{corr} = \frac{\sum_{n=1}^{\infty} nP(n, \mu)}{\sum_{n=1}^{\infty} P(n, \mu)}. \tag{3}$$

Finding the mapping from R^{raw}_{MBD} to R_{true} is simplest to do in reverse, i.e., calculate the mapping from R_{true} to R^{raw}_{MBD}. A lookup table from R^{raw}_{MBD} to R_{true} is created by calculating R^{raw}_{MBD} for many closely spaced values of R_{true}, and is then used to find F_{corr}.

Calculating R^{raw}_{MBD} from R_{true} requires knowing the probability for a bunch crossing to fire both sides of the MBD given n collisions in that bunch crossing. This quantity is written as $P(MBD|n)$ and is determined via PYTHIA-8 events run through GEANT-4 simulation. Tables 1 and 2 show the probabilities for the one and two collision cases to fire the MBD in different ways, respectively. Figure 2 shows these probabilities for the one collision case as a function of truth z-vertex in simulation.

Fire both	Fire N	Fire S	Fire neither
0.562	0.175	0.175	0.088

Table 1: Probabilities from PYTHIA-8+ GEANT-4 for a single pp collision to fire (or not fire).

Figure 1 shows a graphical representation of this mapping between MBD rate and true rates.

		Collision 1			
		Fire both	Fire N	Fire S	Fire neither
	Fire both	0.316*	0.098*	0.098*	0.050*
Collision 2	Fire N	0.098*	0.031	0.031*	0.015
	Fire S	0.098*	0.031*	0.031	0.015
	Fire neither	0.050*	0.015	0.015	0.008

Table 2: Probabilities of all possible cases for a two collision beam crossing, where the possibilities for collision one are laid out in the horizontal direction, and the possibilities for collision two are laid out vertically. As an example, the box corresponding to "Fire N" for collision one and "Fire S" for collision two reports the probability from PYTHIA-8 for collision one to fire only the North side of the MBD, and for collision two to fire only the South side of the MBD. The starred boxes are those which will result in a **MBDN&S $\geq$ 1** trigger.

One can then express answer for R^{raw}_{MBD} as

$$R^{raw}_{MBD} = R_{true} \sum_{n=0}^{\infty} P(MBD|n)P(n,\mu) \tag{4}$$

Once the correction factor is calculated, it can be applied to the uncorrected integrated luminosity to find the corrected integrated luminosity as

$$\mathscr{L}_{\text{int}} = F_{corr}\mathscr{L}_{\text{int}}(uncorrected) \tag{5}$$

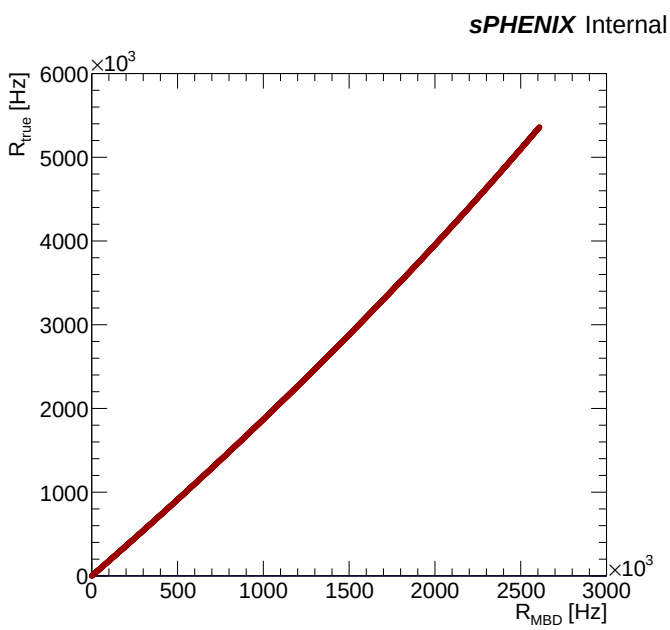

Figure 1: R_{true} corresponding to specific values of R^{raw}_{MBD} as calculated using the method in Subsection 1.2.

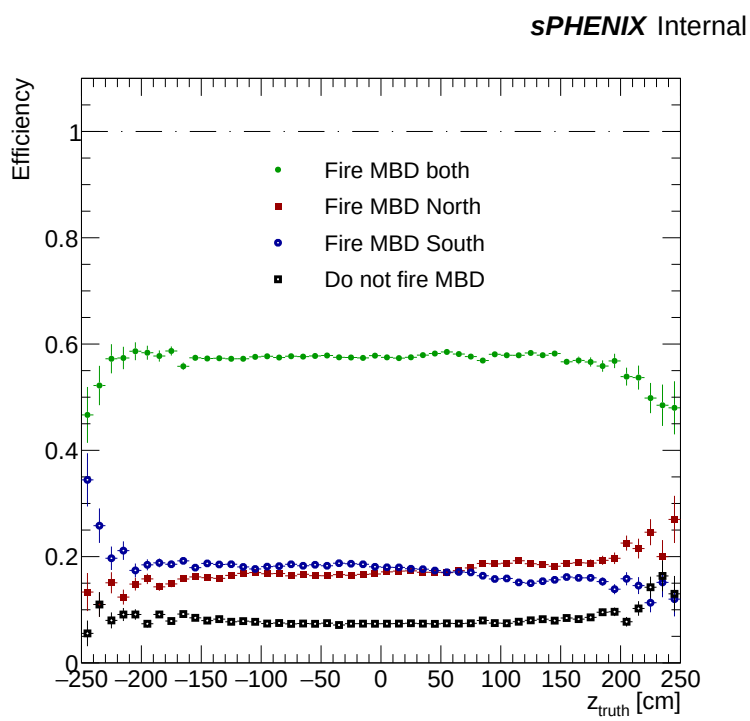

Figure 2: Monte Carlo PYTHIA-8 + GEANT-4 calculated MBD efficiency for the four cases described in Table 1 as a function of truth collision z_{vtx}.

1.3 Calculating the Luminosity

The fully corrected integrated luminosity is then equal to

$$\mathscr{L}_{\text{int}} = \frac{N_{mbd}^{live} \cdot F_{corr}}{P_{Jet10} \cdot \sigma_{MBD}} \tag{6}$$

where N_{mbd}^{live} is the number of live **MBDN&S $\geq$ 1** triggers in the segments in the GRL, P_{Jet10} is the average prescale of the Jet10 trigger, and σ_{MBD} is the MBD cross section – 25.2 mb which is determined from the sPHENIX Vernier Scan [Reference?].

Calculated luminosity values for the minimum bias coincidence, MB coincidence + JET10, and JET10 alone triggers are available at:

https://github.com/sPHENIX-Collaboration/analysis/
blob/master/LuminosityCounterGoodRuns/run/allzlumi.list

Note that for all of these cases, there is no z-vertex selection. Note also that the "MB coincidence + JET10 trigger" luminosity is only different from the "JET10 alone "trigger because sometimes the prescales of the two triggers were not identical. The "minimum bias coincidence only" luminosity is equal to

$$\frac{N_{mbd}^{scaled} \cdot F_{corr}}{\sigma_{MBD}} \tag{7}$$

where the numerator is the number of scaled (not live) counts.

When using the JET10 alone trigger, as in PPG09, all jets are counted, and so there is no correction to the number of jets; all corrections are to the luminosity; however, in any case involving a minimum bias coincidence selection on the trigger, there is an additional correction on the number of jets due to the fact that the trigger is more likely to fire in the double interaction case.

2 QA on Run-to-Run variations

As a check on the run-to-run luminosity calculation, Figures 3, 4, 5 show the number of jets N_{jet} per uncorrected and then corrected luminosity as a function of run number. Figure 6 shows the 1-dimensional distribution of $N_{jet}/\mathscr{L}_{\text{int}}$ values over the entirety of Run 2025, and separated into different regions.
The number of jets is defined as all jets passing the dijet and timing cuts ($\Delta\phi_{l,sl} > 3\pi/4$, $|t_{lead} + 2 \text{ ns}| < 6$ ns, $|\Delta t_{l,sl}| < 3$ ns) with uncalibrated $p_T > 15$ GeV. A 0th order polynomial is fit to this distribution of $N_{jet}/\mathscr{L}_{\text{int}}$ and plotted.
The double interaction correction is only significant during the running period at zero crossing angle (corresponding to runs 51400-53864). There are some remaining non-statistical deviations in the ratio.
Most probably these deviations are related to the jet selection and jet energy calibration. For the final jet cross section measurement, calculating separate regions for the 0 mrad and

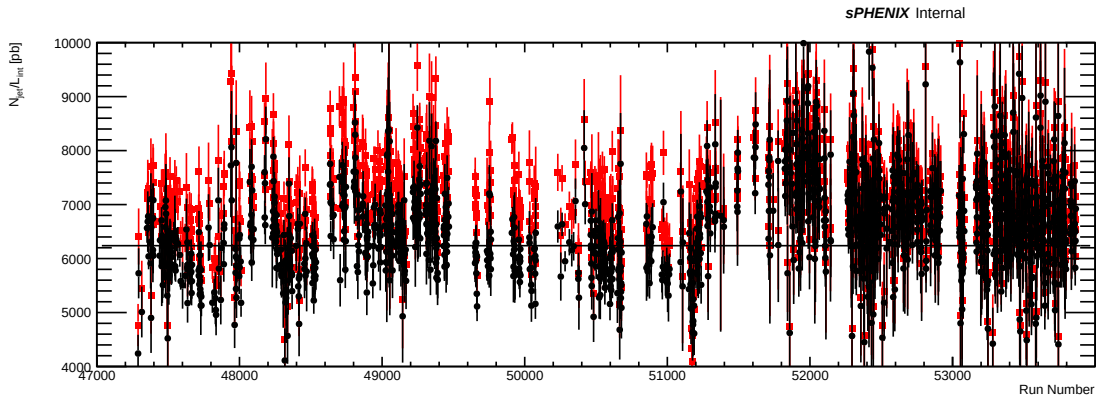

Figure 3: $N_{jet}/\mathscr{L}_{int}$ as a function of run number over the course of Run 2025. The solid black line is a 0th order polynomial fit to these data over the entire range.

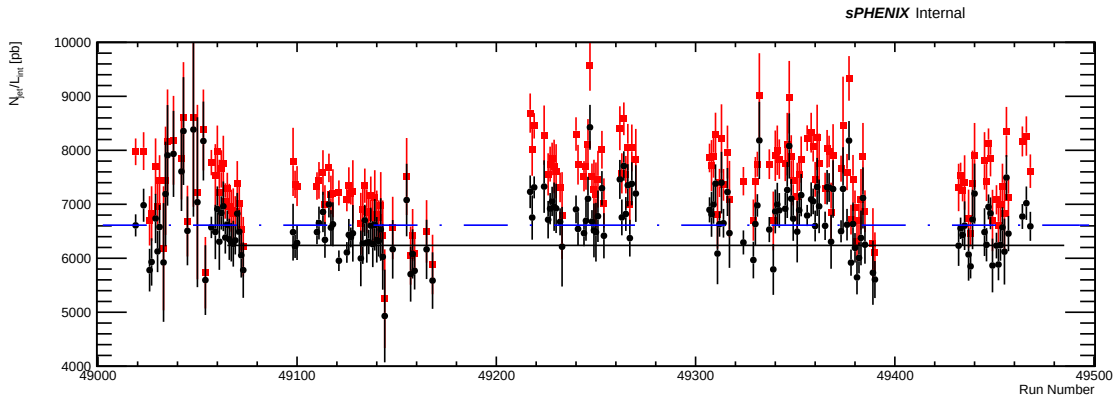

Figure 4: $N_{jet}/\mathscr{L}_{int}$ as a function of run number over the course of Run 2025, zoomed in to a section of 0 mrad runs from run 49000 to run 49500. The solid black line is a 0th order polynomial fit over the entire range of Run 2025, while the blue dashed line is a 0th order polynomial fit to the range on the plot.

1.5 mrad and comparing them at lower pt, i.e., where statistics are very good, might lead to a systematic uncertainty.

There are notable differences in the 1.5 and 0 mrad mean values in Figure 6. To examine this, using the Jet 15 GeV simulation sample, the number of jets with $p_T > 15$ GeV per event in events with a truth z-vertex within 10 cm was compared to the same quantity in events with a truth z-vertex within 50 cm. In data, the size of the effect is roughly 10%, while in simulation, there is roughly a 15% difference, meaning that the two are roughly the same order of magnitude.

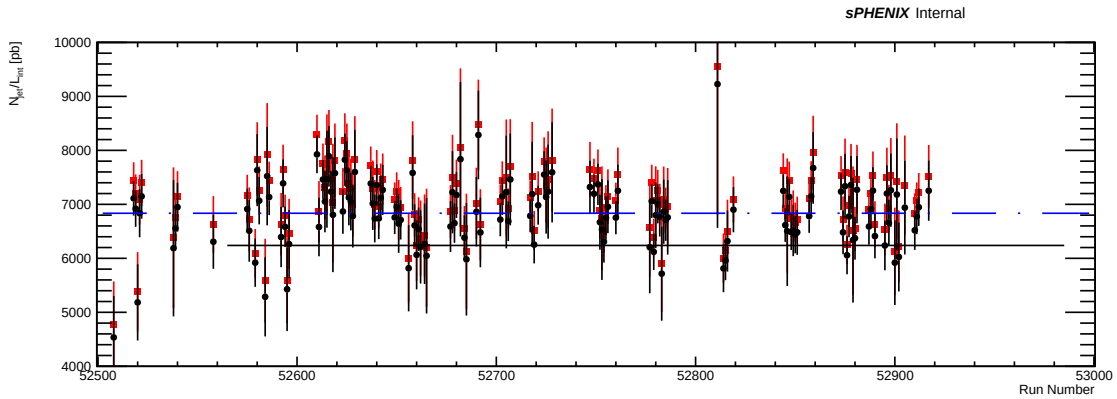

Figure 5: $N_{jet}/\mathscr{L}_{int}$ as a function of run number over the course of Run 2025, zoomed in to a section of 1.5 mrad runs from run 52500 to run 53000. The pileup correction is almost nonexistent for 1.5 mrad running because of the lower rate/collision probability. The solid black line is a 0th order polynomial fit over the entire range of Run 2025, while the blue dashed line is a 0th order polynomial fit to the range on the plot.

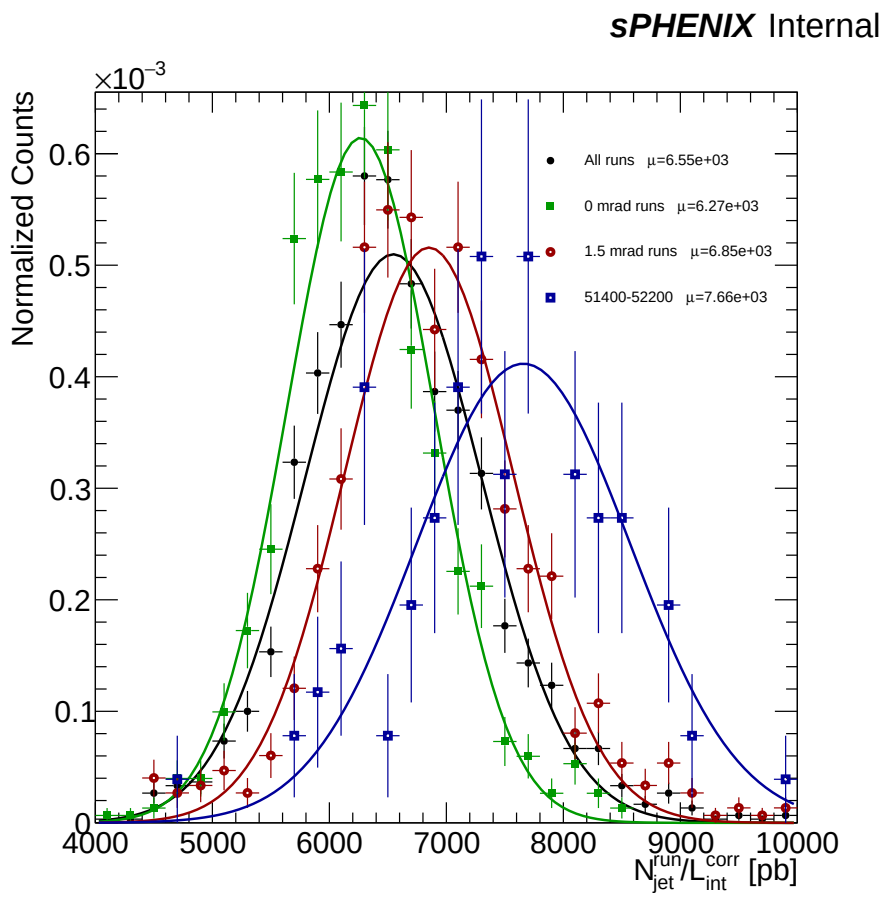

Figure 6: The self-normalized distribution of $N_{jet}/\mathscr{L}_{int}$ for all runs on the GRL. Specific selections are made to separately show 0 mrad running, 1.5 mrad running, and a selection of runs between 51400 and 52200 which have a higher average $N_{jet}/\mathscr{L}_{\text{int}}$.

3 Mapping out the "MBD z-vertex selection" case

In the sections above, the rate of particles is being calculated without requiring an MBD coincidence (**MBDN&S $\geq$ 1**) and without a z-vertex cut. Thus, in a yield calculation, the numerator, e.g., the number of jets, does not require any multiple interaction correction. The denominator has the correction for pile-up as encoded in Figure 1.

However, if the rate of particles is being calculated where the particle counts only include cases where the event has an MBD coincidence and/or an online/offline z-vertex selection, that changes the calculus. In principle, one can leave the denominator calculation identical to the above and just correct the numerator to also account for losses incurred by the z-vertex selection.

Using direct photons as an example, in each segment there is a certain number of direct photon counts in a given p_T bin. As defined above, for a given segment one has a particular rate R of collisions (corresponding to 42 mb interactions). One can then calculate the Poisson probability of there being 1 collision in a bunch crossing, 2 collisions, 3 collisions, etc. – Prob1, Prob2, Prob3, etc. The ROOT method TMath::Poisson() is useful.

One then uses pythia-8events of two types to complete the calculations – pythia-8(γ) and pythia-8(mb) events. The z-vertex distribution of the pythia-8 events **must** match the experimental data for that segment of running.

Using a single pythia-8(γ) event, one can calculate the probability that the event satisfies the MBD coincidence and the specific z-vertex cut – referred to as Pass1. Then one can overlay a single pythia-8(γ) event and a second pythia-8(mb) event and calculate the probability that the event satisfies the MBD coincidence and the specific z-vertex cut – called Pass2. Note that sometimes the two events will collaborate to satisfy the MBD coincidence and sometimes they will conspire to have a z-vertex near zero even when neither collision was in the z-vertex range. Similarly, one can calculate the probability for one single pythia-8(γ) event and two additional pythia-8(mb) events – Pass3.

Then one corrects for the photon counts by dividing by the following.

$$\mathrm{f_{corr}} = \frac{1}{1 - Prob_0} \times [\mathrm{Prob_1} \times \mathrm{Pass_1} + \mathrm{Prob_2} \times \mathrm{Pass_2} + \mathrm{Prob_3} \times \mathrm{Pass_3} + ...]$$

where 1-Prob0 is the sum of probabilities for one or more collision, i.e., Prob1 + Prob2 + Prob3 + ... = 1 - Prob0.

For a collision rate R = 2 MHz, the If one has an overall collision rate of R – number of collisions per second averaged over all bunches, then the average rate of collisions (μ) in a given crossing is R / f, where f is the bunch crossing frequency = 111 bunches $\times$ 78.196 kHz = 8.86 MHz. Thus, for example, for a 2 MHz collision rate, $\mu = 0.225$. The Poisson probabilities Prob1, Prob2, Prob3, Prob4, Prob5 are then 0.179, 0.020, 0.0015, 8.2e-5, 3.8e-6. Thus, it is most likely sufficient to only calculate the first three terms at any pp luminosity.

One technical issue is that the numerator correction depends on the specific probe being measured (for example the p_T of the direct photon). The other dependence is on the z-vertex distribution for that particular segment – noting that the z-vertex distribution not only changes from store-to-store, but is known to change significantly during a store.